Nemo K Cards · @whatsonkorea

팀 Sona · Fastcampus × NVIDIA Agentic AI Hackathon 2026

English follows Korean

요약. 한국을 여행하는 외국인들이 찾고 싶어하는 행사들을 리서치하고 출처를 확인한 뒤 영어 Instagram 카드 뉴스로 만드는 에이전트 파이프라인입니다. 에이전트는 NVIDIA OpenShell sandbox 안에서만 실행되고, 컨텐츠는 사람이 승인한 뒤에만 게시가 됩니다.

문제. 야놀자 리서치 Insights Vol.41(2026.04.26)의 2023–2025년 영어 Reddit 게시물 분석에서 한국 여행 불편 사례의 27.8%가 디지털 서비스, 16.4%가 관광 정보·안내였습니다(분석된 불편 사례 중 비율). Reddit의 r/seoul과 r/koreatravel에는 "행사 소식을 알려주는 영어 앱이나 웹사이트가 있나요?" 같은 질문이 올라옵니다. 정보는 한국어 페이지에 흩어져 있고 날짜, 장소, 입장 조건이 자주 바뀌며 외국인들이 쉽게 접근하지 못합니다.

해결과 구조. 운영자가 Admin(Next.js)에서 요청합니다. FastAPI 백엔드가 단계마다 OpenShell sandbox를 만들어 Claude Code 에이전트를 실행합니다 (리서치 → 정보 검증 → 영어 문구 → 카드 렌더링 → 검수). 링크 확인과 렌더링은 백엔드가 직접 하며 운영자는 카드와 출처, 검수 결과를 보고 승인하면 sandbox 밖의 백엔드가 Instagram Graph API로 게시합니다. 에이전트는 게시할 수 없습니다.

OpenShell 적용 (policies/openshell/agent-policy.yaml)

경계	구현
실행	단계마다 새 sandbox, 로그 수집 후 삭제
파일	에이전트 지침(/sandbox/agent) read-only, /sandbox/out만 쓰기, Landlock hard_requirement
네트워크	기본 거부. 공공 데이터 API와 행사 페이지만 GET 전용으로 허용, 실행 파일(claude, curl)별 지정
Instagram	graph.instagram.com·graph.facebook.com 읽기 전용, POST·DELETE는 L7에서 거부·기록
자격 증명	모델 키는 provider placeholder로만, api.anthropic.com에만 전송. Instagram·저장소 키는 호스트에만
공통 테스트 폴더	/hackathon/input read-only, /hackathon/output 쓰기, restricted·secrets는 조회·읽기 불가

검증. OpenShell 0.1.2 sandbox에서 실행한 probe 기록이 저장소에 있습니다: graph.instagram.com GET 허용, Instagram media POST 거부, graph.facebook.com DELETE 거부, 정책에 없는 pastebin.com 연결 거부. 10/7 Admin의 Policy Log는 /sandbox/agent/CLAUDE.md 쓰기(fs_denied), pastebin.com 연결과 graph.facebook.com DELETE(policy_denied) 거부를 기록했습니다. 특정 테스트의 관찰이며 모든 prompt injection을 막는다는 보장은 아닙니다.

결과. 10/7 게시 실행에서 만든 카드가 @whatsonkorea에 게시되었습니다(예: instagram.com/p/DeL2WwdGQP5). 시스템은 Brev 호스트에서 실행되었습니다. 카드 이미지는 제작 시안이며 모든 카드가 무인 파이프라인을 거쳤다는 의미는 아닙니다.

다음 단계. 10/7 이후, 여러 기관이 각자의 Instagram 계정으로 쓰는 서비스로 확장하고 있습니다. 기관별 계정과 권한, 기관 자체 Instagram 연결, 유해 콘텐츠의 승인을 막는 Safeguard, 재시작에도 이어지는 작업과 중복 없는 게시를 구현해 스테이징에서 검증 중이며, 다음은 Pilot 기관과의 운영입니다.

Nemo K Cards · @whatsonkorea

Team Sona · Fastcampus × NVIDIA Agentic AI Hackathon 2026 · October 7 finals project

Summary: An agent pipeline that researches the events foreigners travelling in Korea want to find, checks their sources, and turns them into English Instagram card news. Agents run only inside NVIDIA OpenShell sandboxes, and content is published only after a person approves it.

Problem. In Yanolja Research Insights Vol.41 (April 26, 2026), an analysis of English Reddit posts from 2023–2025, 27.8% of Korea travel complaints concerned digital services and 16.4% tourist information and guidance (shares of the analysed complaint cases). On Reddit, r/seoul and r/koreatravel see questions like "What's an English app or website that can update foreigners about the events around the city?" The information is scattered across Korean-language pages, and dates, venues and entry conditions change often, so foreigners struggle to reach it.

Solution and architecture. An operator makes a request in the Admin (Next.js). The FastAPI backend creates an OpenShell sandbox for each stage and runs a Claude Code agent in it (research → verify → English copy → card rendering → review). The backend checks links and renders cards itself. When the operator reviews the cards, sources and check results and approves, the backend, outside the sandbox, publishes through the Instagram Graph API. Agents cannot publish.

How OpenShell is applied (policies/openshell/agent-policy.yaml)

Boundary	Implementation
Execution	A fresh sandbox per stage; logs collected, then deleted
Files	Agent instructions (/sandbox/agent) read-only; only /sandbox/out writable; Landlock hard_requirement
Network	Deny by default. Public data APIs and event pages allowed read-only (GET), bound to binaries (claude, curl)
Instagram	graph.instagram.com and graph.facebook.com read-only; POST and DELETE denied at L7 and logged
Credentials	Model key only as a provider placeholder, sent only to api.anthropic.com; Instagram and storage keys only on the host
Common-test folders	/hackathon/input read-only, /hackathon/output writable; restricted and secrets can be neither listed nor read

Evidence. A probe captured in a real OpenShell 0.1.2 sandbox is in the repository: GET to graph.instagram.com allowed, an Instagram media POST denied, a DELETE to graph.facebook.com denied, and a connection to pastebin.com (in no policy) refused. On October 7 the Admin's Policy Log recorded a denied write to /sandbox/agent/CLAUDE.md (fs_denied) and denied pastebin.com and graph.facebook.com DELETE requests (policy_denied). These are specific test observations, not a guarantee against every prompt injection.

Results. Cards from the October 7 publishing run were posted to @whatsonkorea (for example instagram.com/p/DeL2WwdGQP5). The system ran on a Brev host. The card images are project designs and don't show that every card went through an unattended pipeline.

Next. Since October 7 we have been turning it into a service that institutions run on their own Instagram accounts: organization accounts with roles, connecting an organization's own Instagram, a Safeguard that blocks approval of harmful content, jobs that survive restarts, and publishing that never posts twice. These run on staging today; next is operating it with pilot institutions.

부록 · Appendix: 근거 · Sources

Repository github.com/juyoungml/nemo-k-cards, as of commit a3675d7 (October 7, 2026), unless noted.

Claim	Source
Pipeline (research, verify, copy, render, review), human approval, publishing by the backend only	README.en.md · presentation slide 05
Sandbox per stage; file, network, Instagram and credential boundaries; /hackathon folders	policies/openshell/agent-policy.yaml · README.en.md "Where OpenShell fits"
Model key only as a provider placeholder, sent only to api.anthropic.com	policies/openshell/providers/claude-code.yaml
OpenShell 0.1.2 probe: Instagram GET allowed, media POST denied, Facebook DELETE denied, pastebin.com refused	backend/tests/data/openshell-probe.txt · docs/SHOWCASE.md
October 7 Policy Log (fs_denied, policy_denied) and its scope	Presentation slide 08 · docs/SHOWCASE.md "Policy evidence"
Yanolja Research Insights Vol.41 figures and scope; Reddit questions	Presentation slides 02–03
Architecture on a Brev host; card images are designs	Presentation slides 01, 09
October 7 posts on @whatsonkorea	README.en.md "See the result"
Work since October 7 (Next)	Merged pull requests #36–#50 on main